

Water, minerals and electrolytes

First stage

2026



Water

Importance of water

- Essential constituent of all forms of life
- Required for transport of solutes in the body
- Required for enzyme action
- Regulation of body temperature
- Providing ions for accelerating biochemical reactions.

Distribution of water in the body

Total body water

42L

60% of the body weight

major compartments



Intracellular fluid

28 L

40% of the body weight

Extracellular fluid

14 L

20% of the body

Factors affecting water balance

- Rate of metabolism
- Urine volume
- Osmolality of blood
- Climate.

Regulation of body water

Body water deficient-dehydration

Increase plasma osmolality



Stimulates the thirst center
in the hypothalamus

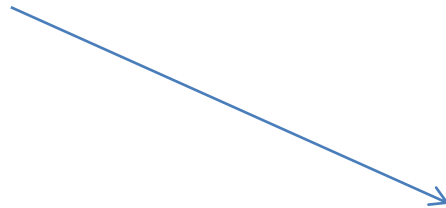
Thirst

stimulates receptors in
the posterior pituitary
to release anti diuretic
hormone (ADH)



Increase intake of water

increase reabsorption of
water from collecting ducts



Increase body water back to normal

Table(1) regulation of water balance

Daily input	Daily output
Drinking 1000ml	Urine 1000-1500ml
Eating 700-1000 ml	Sweat 400 ml
Water produced by metabolism 300-500ml	Respiration 400ml
	Faeces 200ml
Total 2000-2500ml	Total 2000-2500 ml

Disturbance of water balance:

Dehydration (Hypovolemia):

Loss of water from the body in excess amounts leads to dehydration.

First the plasma becomes concentrated followed by other ECF and later ICF.

When water comes out of the cells, it also passes into ECF in exchange of K & Na passes into ICF from ECF. Loss of more than 20% of body water results in death.

Causes of dehydration:

- 1-Severe diarrhea and vomiting
- 2-Nonavailability of drinking water as in the case of desert.
- 3-Difficulty in swallowing and state of unconsciousness.
- 4-Loss of fluid from skin in case of burns.
- 5-Diabetic insipidus (ADH deficiency-polyuria)
- 6-Heart stroke

Features of dehydration:

- 1-Dryness of skin, tongue, throat
- 2-Changes in the values of packed cell volume, Hb, plasma protein, plasma electrolytes, urea and decreased blood volume.

Treatment:

Consuming plenty of plain water or water containing sugar and salt depends upon the cause of dehydration. if the condition is very severe, intravenous infusion of fluids (normal saline) is required.

Water excess (hydration) hypervolemia:

The condition in which body water content is increased.

Causes:

- 1-Hypersecretion of ADH
- 2- Following the administration of anesthetics his effect takes for about 12-36 hour after the surgery.
- 3-Renal failure.

Features:

- 1-Mental confusion, in coordination, muscular weakness, nausea.
- 2-Dereased PCV.
- 3-Dereased plasma electrolytes, plasma osmolality, Increased urine osmolality and increased blood pressure.

Treatment:

- 1-Withdrawal of fluids.
- 2-Administration of diuretics.

Minerals and electrolytes

Minerals

Minerals are inorganic ions and elements required with the diet for growth, maintenance of tissue and regulation of body processes.

Minerals may be divided into two groups;

1-Macrominerals

2-Microminerals

1-macrominerals

Which are required in amounts in greater than 100mg/day, which include;
(Calcium, Phosphorus, Sodium, Potassium, Chloride, and Magnesium).

2-microminerals

Which are required in amounts in less than 100mg/day, which include; (Iron, Iodine, Manganese, Cobalt, Fluoride, Chromium, Selenium, Zinc and, Copper).

Macrominerals

Sodium; [Na]

Is the major cation of the extra cellular fluid (ECF).

Sources;

Milk, cereal, legumes, eggs, vegetables like carrots and tomato, and table salt.

Functions:

1. responsible for osmolality of the plasma.
2. plays a main role in maintaining the normal distribution of water and the osmotic pressure in the ECF compartment.
3. regulate the electrolyte and pH balance of the extra cellular compartment.
4. helps in the active transport of **glucose, galactose, and amino acids** across intestinal mucosa and for **Na⁺/K⁺ ATPase**.

Hyponatremia;

Decrease in plasma concentration of [Na ion] may be due to defect in kidneys or adrenal cortex. sweating, burns, vomiting, or diarrhea which can be cause loss of [Na ion] containing fluids.

In adrenal cortex insufficiency

(Addison's disease), decrease of serum [Na ion] and increase in Na ion secretion are seen.

Hypernatremia;

Increased plasma concentration of [Na ion]. it is due to decrease loss of [Na ion] in body fluids, may occur in decrease activity of ADH, Hyperactivity of the adrenal cortex as in Cushing's syndrome. Administration of cortisone also leads to increase in serum Na ion.

Hypertensive patients are advised to take less salt. about 1gm/day.

potassium; [K ion]

Is the major cation of the intra cellular fluid (ICF).

Sources;

vegetables, fruits, and nuts

Functions;

- 1-Is the principle cation of ICF.
- 2-[K] ion is required for functioning of nerve, skeletal muscle and cardiac muscles..
- 3-[K] ion is involved in acid-base balance.

Hypokalemia;

Decrease in plasma concentration of [K ion] This condition decreases the heart beat and interferes with vital muscles as those involved in respiration. Hypokalemia can occur in any illness, wasting disease, intestinal fistulas and in diarrhea.

Since aldosterone increases the excretion of K ion. Cushing syndrome or administration of cortisone leads to hypokalemia. certain diuretics increase the excretion of K ion.

Therefore, important to supplement enough K ion when these diuretics are used.

Hyperkalemia;

Increased plasma concentration of [K ion]. it is occurs in Addison's disease and in intra venous infusion of K ion at a rate excess of 25 mmol/hour.

Causes are acute renal failure, chronic renal failure, glomerulonephritis.

Tissue trauma causes the cells to release K ion into the ECF are burns, traumatic injury and intestinal bleeding.

Chloride and Bicarbonate ions

Chloride ion

- Major extracellular anion

Regulation

- Chloride absorbed in intestinal tract.
- Filtered out by glomerulus.
- Counter movement with sodium.

Maintenance

Chloride shift (sodium reabsorption is limited by amount of chloride by counter current).

Significance of body's chloride

- One of the most important of electrically charged minerals is chloride.
- It works with other electrolytes, such as sodium and potassium, to help balance acids and bases in our body.
- It also helps move fluid in and out of body cells.
- It also helps maintain blood volume, blood pressure and PH of your body's fluid.

Sources of chloride

- Chloride is found in table salt or sea salt as sodium chloride.
- It is also found in many vegetables.
- Foods with higher amounts of chloride include seaweed, rye, tomatoes, lettuce, celery and olives.
- Chloride combined with potassium.

Hypochloremia

- **Addison's disease**

Aldosterone decreases, reabsorption of chloride decreases and concentration of chloride ion decreases

- **Respiratory alkalosis**

Elimination of CO₂ increase, blood bicarbonate concentration increases and compensatory chloride ion concentration decrease

Hyperchloremia

- **Dehydration**
- **Cushing's syndrome**
reabsorption of chloride at renal tubules
increases
- **Sever diarrhea**, loss of bicarbonate and
compensatory retention of chloride.

Bicarbonate ion

Second most abundant anion in the ECF.

Alkalosis:

- Increased HCO_3^- hence increased CO_2 .
- Secretion of HCO_3^- increased in urine.

Acidosis:

- Increased excretion of H^+ into urine.
- CO_2 decreases in blood.

Anion gap

The sum of cations and anions in extracellular fluid is always equal so as to maintain the electrical neutrality. Cations (95%) were maintained by (Na and K ions), whereas, (Cl⁻ and HCO₃⁻ ions), account for (86%) of anions. There are commonly measured electrolytes and hence there is a difference between cations and anions.

The difference of cations and anions or the unmeasured anions constitute the anion gap, which is due to phosphorous, SO_4 ion, PO_4 ion, and organic salts.

The difference between (Na +K ions) and (Cl + HCO_3 ions) is normally about $12 \pm 5 \text{ mEq/L}$ (mMols/L). Alteration in anion gap is extremely useful in the clinical assessment with acid-base disorders.

$$\text{Anion gap} = \{[\text{Na}] + [\text{K}]\} - \{[\text{Cl}] + [\text{HCO}_3]\}$$